

Exercise 1: The Projector

We defined the projector $h^{\mu\nu}$ on $\Sigma \subset \mathcal{M}$ as $h^{\mu\nu} = h^{ij} e_i^\mu e_j^\nu$, where $e_i^\mu = \frac{\partial x^\mu}{\partial y^i}$, fulfilling

$$g^{\mu\nu} = h^{\mu\nu} + \varepsilon n^\mu n^\nu, \quad (1)$$

where n_μ is the unit normal to Σ , with $\varepsilon = n^2$, fulfilling $e_i^\mu n_\mu = 0$ for all i .

a)* Prove

$$h_\nu^\mu = \delta_\nu^\mu - \varepsilon n^\mu n_\nu, \quad (2)$$

$$h^{\mu\nu} n_\mu = 0, \quad (3)$$

$$h_\rho^\mu h_\nu^\rho = h_\nu^\mu. \quad (4)$$

b) We also defined $e_\mu^i = h^{ij} g_{\mu\nu} e_j^\nu$. Prove

$$e_i^j e_j^\mu = \delta_i^\mu, \quad (5)$$

$$e_\mu^i e_j^\mu = h_\nu^i, \quad (6)$$

$$e_\mu^i n_\mu = 0. \quad (7)$$

$$\begin{aligned} h_\gamma^\nu &= g_{\gamma\sigma} h^{\nu\sigma} \\ &= g_{\gamma\sigma} (g^{\nu\sigma} - \varepsilon n^\sigma n^\gamma) \\ &= \delta_\gamma^\nu - \varepsilon n^\nu n_\gamma \\ h^{\nu\gamma} n_\gamma &= (g^{\nu\gamma} - \varepsilon n^\gamma n^\nu) n_\gamma \\ &= n^\nu - \varepsilon n^\nu n^2 \\ &= n^\nu - (n^2)^2 n^\nu = 0 \\ h_\rho^\mu h_\gamma^\rho &= (\delta_\rho^\mu - \varepsilon n^\mu n_\rho) (\delta_\gamma^\rho - \varepsilon n^\rho n_\gamma) \\ &= \delta_\rho^\mu \delta_\gamma^\rho - \varepsilon \delta_\rho^\mu n^\rho n_\gamma - \varepsilon \delta_\gamma^\rho n^\mu n_\rho + \varepsilon n^\mu n_\rho n^\rho n_\gamma \\ &= \delta_\gamma^\mu - \varepsilon n^\mu n_\gamma - \varepsilon n^\mu n_\gamma + \varepsilon n^\mu n_\gamma \\ &= \delta_\gamma^\mu - \varepsilon n^\mu n_\gamma = h_\gamma^\mu \end{aligned}$$

$$\begin{aligned} b) \quad e_i^\mu e_j^\nu &= (h^{ik} g_{\mu\nu} e_k^\nu) e_j^\mu \\ &= h^{ik} h_{kj} = \delta_i^j \\ e_i^\mu e_j^\nu &= (h^{ij} g_{\mu\nu} e_j^\nu) e_i^\mu \\ &= h^{\gamma\gamma} g_{\mu\nu} \\ &= h_\mu^\nu \\ e_i^\mu n_\mu &= (h^{ij} g_{\mu\nu} e_j^\nu) n_\mu \\ &= h^{ij} e_j^\nu n_\nu \\ &= 0 \end{aligned}$$

Exercise 2: Extrinsic Curvature

We defined the extrinsic curvature as $K_{\mu\nu} = -h_\mu^\lambda \nabla_\lambda n_\nu$.

a)* Prove

$$K_{\mu\nu} = -h_\mu^\lambda h_\nu^\sigma \nabla_\lambda n_\sigma, \quad (8)$$

$$K_{\mu\nu} = n_\sigma h_\mu^\rho \nabla_\rho h_\nu^\sigma. \quad (9)$$

$$\begin{aligned} K_{\nu\sigma} &= -h_\nu^\lambda \nabla_\lambda n_\sigma \\ \delta_\gamma^\sigma &= h_\gamma^\sigma + \varepsilon n^\sigma n_\gamma \\ K_{\nu\gamma} &= -h_\nu^\lambda \delta_\gamma^\sigma \nabla_\lambda n_\sigma \\ &= -h_\nu^\lambda (h_\gamma^\sigma + \varepsilon n^\sigma n_\gamma) \nabla_\lambda n_\sigma \\ \text{But } n^\sigma \nabla_\lambda n_\sigma &= \frac{1}{2} \nabla_\lambda (n^\sigma n_\sigma) = 0, \text{ thus the second term vanishes} \\ K_{\nu\gamma} &= -h_\nu^\lambda h_\gamma^\sigma \nabla_\lambda n_\sigma \\ K_{\nu\gamma} &= -h_\nu^\lambda \nabla_\lambda n_\gamma \\ &= -h_\nu^\lambda \nabla_\lambda (\delta_\gamma^\sigma n_\sigma) \end{aligned}$$

$$\begin{aligned}
&= -h^\lambda{}_\mu \nabla_\lambda \left[(h^\sigma{}_\nu + \varepsilon n^\sigma n_\nu) n_\sigma \right] \\
&= -h^\lambda{}_\mu \nabla_\lambda \left[h^\sigma{}_\nu n_\sigma + n_\nu \right] \\
&= -h^\lambda{}_\mu \left[h^\sigma{}_\nu \nabla_\lambda n_\sigma + n_\sigma \nabla_\lambda h^\sigma{}_\nu + \nabla_\lambda n_\nu \right]
\end{aligned}$$

We see that

$$\nabla_\lambda n_\nu = h^\sigma{}_\nu \nabla_\lambda n_\sigma + n_\sigma \nabla_\lambda h^\sigma{}_\nu + \nabla_\lambda n_\nu$$

$$0 = h^\sigma{}_\nu \nabla_\lambda n_\sigma + n_\sigma \nabla_\lambda h^\sigma{}_\nu$$

$$\begin{aligned}
\text{Hence } K_{\nu\gamma} &= -h^\lambda{}_\nu h^\sigma{}_\gamma \nabla_\lambda n_\sigma \\
&= h^\lambda{}_\nu n_\sigma \nabla_\lambda n^\sigma{}_\gamma
\end{aligned}$$

b) Writing $n = N df$, with $N = [g^{\mu\nu} \partial_\mu f \partial_\nu f]^{-1/2}$, prove that $K_{\mu\nu}$ is symmetric.

$$n_\nu = N \partial_\nu f$$

$$\begin{aligned}
K_{\nu\gamma} &= -h^\lambda{}_\nu h^\sigma{}_\gamma \nabla_\lambda n_\sigma \\
&= -h^\lambda{}_\nu h^\sigma{}_\gamma \nabla_\lambda (N \partial_\sigma f) \\
&= -h^\lambda{}_\nu h^\sigma{}_\gamma [\partial_\lambda N \partial_\sigma f + N \nabla_\lambda \partial_\sigma f]
\end{aligned}$$

$$\text{Since } h^\sigma{}_\nu \partial_\sigma f \propto h^\sigma{}_\nu n_\sigma = 0,$$

$$K_{\nu\gamma} = -h^\lambda{}_\nu h^\sigma{}_\gamma N \nabla_\lambda \partial_\sigma f$$

This is manifestly symmetric.

c)* Assuming that $K_{\mu\nu}$ is symmetric, prove

$$K_{\mu\nu} = -\frac{1}{2} \mathcal{L}_n h_{\mu\nu}. \quad (10)$$

Interlude: Lie derivative of a $(0,2)$ tensor.

$$(\mathcal{L}_X t)(v_1, \dots, v_r) = X(t(v_1, \dots, v_r)) - \sum_{i=1}^r t(v_1, \dots, [X, v_i], \dots, v_r)$$

In coordinates

$$\begin{aligned}
(\mathcal{L}_n h)(v_i, v_j) &= n(h(v_i, v_j)) - h(v_i, [n, v_j]) - h([n, v_i], v_j) \\
&= n^\rho \partial_\rho (h_{\gamma\rho} v_i^\gamma v_j^\rho) - h_{\gamma\rho} v_i^\gamma (n^\rho \partial_\rho v_j^\rho - v_j^\rho \partial_\rho n^\rho) - h_{\gamma\rho} (n^\rho \partial_\rho v_i^\rho - v_i^\rho \partial_\rho n^\rho) v_j^\rho
\end{aligned}$$

Taking the v s to be the unit vectors, we find

$$(\mathcal{L}_n h)_{\mu\nu} = n^\rho (\partial_\rho h_{\mu\nu}) + h_{\rho\mu} (\partial_\rho n^\rho) + h_{\rho\nu} (\partial_\rho n^\rho)$$

In this exercise we will perform some necessary calculations for the Oppenheimer-Snyder collapse, which we will study in detail in class on Wednesday, May 13th.

In 1939, J. Robert Oppenheimer and his student Hartland Snyder published the first solution to the Einstein Equations describing a star's dynamical collapse to form a black hole. They crudely modelled the collapsing star as a spherical ball of pressureless matter of uniform density.

Therefore, the metric inside the star is a FLRW metric with $k = 1$,

$$ds^2 = -d\tau^2 + a^2(\tau) [d\chi^2 + \sin^2(\chi) d\Omega^2], \quad (11)$$

while the metric outside the star, due to Birkhoff's theorem, must be the Schwarzschild metric,

$$ds^2 = -\left(1 - \frac{2M}{r}\right) dt^2 + \frac{dr^2}{1 - \frac{2M}{r}} + r^2 d\Omega^2, \quad (12)$$

where M is the star's total mass. Notice that the time coordinates t and τ are not the same.

To glue these two metrics along the surface of the star, we must first compute the induced metric and extrinsic curvature on the surface of the star, as seen from the two perspectives.

From the inside, the surface of the star is located at constant $\chi = \chi_0$ in comoving coordinates.

From the outside, the surface of the star is at some $r = R(\tau)$ and $t = T(\tau)$, where τ is the proper time of observers at the surface of the star. Clearly this τ is the same as in (11).

a)* Choosing $y^i = (\tau, \theta, \phi)$ as coordinates on the surface of the star Σ , show that the induced metric, as seen from the inside, reads

$$ds^2|_{\Sigma} = -d\tau^2 + a^2(\tau) \sin^2 \chi_0 d\Omega^2, \quad (13)$$

while, as seen from outside, it is

$$ds^2|_{\Sigma} = -\left(F\dot{T}^2 - F^{-1}\dot{R}^2\right) d\tau^2 + R^2 d\Omega^2, \quad (14)$$

where $F = 1 - 2M/R$. Hint:

Since the induced metrics must match, this means

$$R(\tau) = a(\tau) \sin \chi_0, \quad F\dot{T}^2 - F^{-1}\dot{R}^2 = 1. \quad (15)$$

a) We basically only touch $\chi = \chi_0$, the rest stay untouched. Thus

$$ds^2|_{\Sigma} = -d\tau^2 + a(\tau)^2 \sin^2 \chi_0 d\Omega^2$$

From outside, we substitute $r = R(\tau)$, $t = T(\tau)$

$$\begin{aligned} ds^2 &= -\left(1 - \frac{2M}{R}\right) \dot{T}^2 d\tau^2 + \frac{\dot{R}^2 d\tau^2}{1 - \frac{2M}{R}} + R^2 d\Omega^2 \\ &= -(F\dot{T}^2 - F^{-1}\dot{R}^2) d\tau^2 + R^2 d\Omega^2 \end{aligned}$$

b)* Show that the unit normal vector, as seen from the inside, is

$$n^\mu = a^{-1} \delta^\mu_\tau, \quad (16)$$

Since Σ is determined by $\chi = \chi_0$, $n \parallel d\chi$

$$n^\nu = g^{\nu\mu} n_\mu = \frac{1}{a} n_\nu \delta^\nu_\chi$$

$$n^\nu n_\nu = 1, \text{ thus}$$

$$n^\nu = \frac{1}{a} \delta^\nu_\chi$$

c)* Compute the extrinsic curvature, as seen from the inside,

$$K_{\tau\tau} = 0, \quad K_{\theta\theta} = -a \sin \chi_0 \cos \chi_0, \quad K_{\phi\phi} = -a \sin \chi_0 \cos \chi_0 \sin^2 \theta. \quad (17)$$

We use the Lie derivative formula as in the hint

$$\begin{aligned} K_{\mu\nu} &= -\frac{1}{2} \mathcal{L}_n h_{\mu\nu} \\ &= n^\rho (\partial_\rho h_{\mu\nu}) + h_{\rho\nu} (\partial_\mu n^\rho) + h_{\rho\mu} (\partial_\nu n^\rho) \end{aligned}$$

Since $h_{\chi\chi} = \text{const}$, its Lie derivative vanishes

$$K_{\chi\chi} = 0$$

We have $h_{\theta\theta} = a^2 \sin^2 \chi$ and since $n \parallel d\chi$ the last 2 terms vanish

$$\begin{aligned} n^\rho \partial_\rho h_{\theta\theta} &= \frac{1}{a} \partial_\chi (a^2 \sin^2 \chi) \\ &= 2a \sin \chi \cos \chi \end{aligned}$$

$$\begin{aligned} K_{\theta\theta} &= -\frac{1}{2} n^\rho \partial_\rho h_{\theta\theta} |_{\chi=\chi_0} \\ &= -a \sin \chi_0 \cos \chi_0 \end{aligned}$$

Similarly, $h_{\phi\phi} = a^2 \sin^2 \theta \sin^2 \chi$,

$$\begin{aligned} K_{\theta\phi} &= -\frac{1}{2} n^\rho \partial_\rho (a^2 \sin^2 \theta \sin^2 \chi) |_{\chi=\chi_0} \\ &= -a \sin \chi_0 \cos \chi_0 \sin^2 \theta \end{aligned}$$

d)* Show that the unit normal covector, as seen from the outside, is

$$n_\mu = (\dot{R}, -\dot{T}, 0, 0). \quad (18)$$

Again these are determined by $r = R(\tau)$, $t = T(\tau)$

It can be seen by direct computation that $(\dot{R}, -\dot{T}, 0, 0)$ is a normal vector.

Normalisation: Define $n_\nu = (\dot{R}, -\dot{T}, 0, 0)$

$$n^\nu = (-F\dot{R}, -F\dot{T}, 0, 0)$$

$$n^\nu n_\nu = -F\dot{R}^2 + F\dot{T}^2 = 1 \quad (\text{eq. 15})$$

Hence $\vec{n}_p$ is already normalized and works as the normal vector.

In a D -dimensional manifold M , consider a hypersurface $\Sigma = \{r(x^\mu) = 0\}$. Gaussian normal coordinates on M adapted to this hypersurface are defined by the conditions that $g^{\mu\nu} \partial_\mu r \partial_\nu r = 1$ and $g^{\mu\nu} \partial_\mu r \partial_\nu y^i = 0$, where the y^i are coordinates on Σ . It can be shown that these conditions always locally admit a solution.

a) Show that if we choose (r, y^i) as Gaussian normal coordinates on a neighbourhood of $\Sigma \subset M$, then the metric can locally be written as

$$ds^2 = dr^2 + h_{ij}(r, y^k) dy^i dy^j. \quad (21)$$

We compute the components of the metric tensor

$$\begin{aligned} g^{rr} &= g(\partial_r, \partial_r) \\ &= g((\partial_\mu r) \partial_\mu, (\partial_\nu r) \partial_\nu) \\ &= g^{\mu\nu} \partial_\mu r \partial_\nu r = 1 \end{aligned}$$

$$g^{ri} = g^{\mu\nu} \partial_\mu r \partial_\nu y^i = 0$$

This eliminates terms like $dr dy^i$ and show that the coefficient of dr^2 is 1

The rest is just the most general possible form.

b) Find the expression for the extrinsic curvature of Σ in these coordinates.

$$\begin{aligned} K_{\mu\nu} &= -\frac{1}{2} \mathcal{L}_n h_{\mu\nu} \\ &= -\frac{1}{2} \partial_r h_{\mu\nu} \end{aligned}$$

(normal vector is "constant" in the partial derivative sense along the surface)

Exercise 5: ADM Decomposition

Consider a metric written in the $3+1$ form,

$$ds^2 = -N^2 dt^2 + h_{ij} (dx^i + N^i dt) (dx^j + N^j dt). \quad (22)$$

Let $T = \partial_t$, $X = N^i \partial_i$, and $n = (T - X)/N$.

a) Verify that $n_\mu = -N(dt)_\mu$ and hence n^μ is the future-directed unit normal vector to $\Sigma_t = \{t = \text{constant}\}$.

$$\begin{aligned} n &= \frac{1}{N} (\partial_t - N^i \partial_i) \\ n^0 &= \frac{1}{N}, \quad n^i = -\frac{1}{N} N^i \end{aligned}$$

Now we compute the metric elements

$$g_{00} = -N^2 + h_{ij} N^i N^j, \quad g_{0i} = h_{ij} N^j, \quad g_{ij} = h_{ij}$$

$$\begin{aligned} \text{Hence } n_0 &= g_{00} n^0 + g_{0i} n^i \\ &= (-N^2 + h_{ij} N^i N^j) \frac{1}{N} - h_{ij} N^j \frac{1}{N} N^i \\ &= -N \\ n_i &= g_{i\mu} n^\mu \\ &= g_{i0} n^0 + g_{ij} n^j \\ &= h_{ij} N^j \frac{1}{N} + h_{ij} \left(-\frac{1}{N} N^j\right) \\ &= 0 \end{aligned}$$

$$\begin{aligned} n_\mu &= (-N, 0, 0, 0) \\ &= -N(dt)_\mu \end{aligned}$$

b) By writing n_μ in terms of T_μ and X_μ and using $K_{\mu\nu} = h^\alpha_\mu h^\beta_\nu \nabla_\rho n_\alpha$, show that

$$K_{ij} = \frac{1}{2N} (\partial_t h_{ij} - D_i N_j - D_j N_i), \quad (23)$$

where D_i is the Levi-Civita connection of h_{ij} , and $N_i = h_{ij} N^j$.

$$\begin{aligned} K_{ij} &= h_i^\alpha h_j^\beta \nabla_\rho n_\alpha \\ n_\mu &= -N dt \\ \nabla_\rho n &= -(\partial_\rho N) dt - N \nabla_\rho (dt) \\ \partial_\rho n_\sigma &= -(\partial_\rho N) \partial_\sigma t - N \nabla_\rho \nabla_\sigma t \end{aligned}$$

First term vanishes after projecting $h^\sigma_\gamma \partial_\sigma t = 0$

$$\nabla_\rho \pi_\sigma = -N \nabla_\rho \mathcal{L}_\sigma + N \Gamma_{\rho\sigma}^0$$

Hence we need $K_{ij} = N \Gamma_{ij}^0$

$$\Gamma_{ij}^0 = \frac{1}{2} g^{00} (\partial_i g_{0j} + \partial_j g_{0i} - \partial_0 g_{ij}) + \frac{1}{2} g^{0k} (\partial_i g_{kj} + \partial_j g_{ki} - \partial_k g_{ij})$$

$$= -\frac{1}{2N^2} (\partial_i N_j + \partial_j N_i - \partial_0 h_{ij}) + \frac{1}{2} \frac{N^k}{N^2} (\partial_i h_{kj} + \partial_j h_{ki} - \partial_k g_{ij})$$

combining into covariant derivative

$$= \frac{1}{2N^2} (\partial_i h_{ij} - D_i N_j - D_j N_i)$$

Exercise 6: Gauss' Theorema Egregium

Recall that we defined the Riemann tensor $\mathcal{R}^\mu{}_{\nu\rho\sigma}$ of the connection D_μ on $\Sigma \subset \mathcal{M}$ as

$$\mathcal{R}^\mu{}_{\nu\rho\sigma} V^\nu = 2D_{[\rho} D_{\sigma]} V^\mu, \quad (24)$$

for vectors V^μ tangent to $\Sigma \subset \mathcal{M}$, that is, for vectors fulfilling $V^\mu n_\mu = 0$.

a) *Extra exercise (not compulsory):* Prove Gauss' Equation,

$$h^\mu_\alpha h^\beta_\nu h^\gamma_\rho h^\delta_\sigma R^\alpha{}_{\beta\gamma\delta} = \mathcal{R}^\mu{}_{\nu\rho\sigma} - 2\varepsilon K_{[\rho}{}^\mu K_{\sigma]\nu}, \quad (25)$$

Warning: This is not a short computation. However, it is also not difficult. Simply expand the RHS of eq. (24) using the definition of D_μ and identify the relevant terms.

b) Contract Gauss' Equation to find the Contracted Gauss Equation,

$$R - 2\varepsilon n^\mu n^\nu R_{\mu\nu} = \mathcal{R} + \varepsilon(K^{\mu\nu} K_{\mu\nu} - K^2). \quad (26)$$

Hint:

Notice that in both cases, if our ambient spacetime $\mathcal{M}$ is flat, then $R^\alpha{}_{\beta\gamma\delta} = 0$, and so the intrinsic curvature $\mathcal{R}$ of the hypersurface Σ is directly related to its embedding in $\mathcal{M}$, that is, to its extrinsic curvature K .

$$\begin{aligned} h^\mu_\alpha h^\beta_\nu h^\gamma_\rho h^\delta_\sigma R^\alpha{}_{\beta\gamma\delta} &= R^\mu{}_{\nu\rho\sigma} - 2\varepsilon K_{[\rho}{}^\mu K_{\sigma]\nu} \\ &= R^\mu{}_{\nu\rho\sigma} - \varepsilon(K_\rho{}^\mu K_{\sigma\nu} - K_\sigma{}^\mu K_{\rho\nu}) \end{aligned}$$

$$h^\mu_\alpha h^\beta_\nu h^\gamma_\rho h^\delta_\sigma R_{\mu\beta\gamma\delta} = R_{\nu\rho\sigma} - \varepsilon(K_{\rho\mu} K_{\sigma\nu} - K_{\sigma\mu} K_{\rho\nu})$$

$$R_{\nu\sigma} = h^\mu{}_\rho R_{\mu\nu\sigma}$$

$$\begin{aligned} h^\mu{}_\rho h^\alpha{}_\nu h^\beta{}_\gamma h^\delta{}_\sigma R_{\mu\alpha\beta\gamma\delta} &= h^\beta{}_\gamma h^\alpha{}_\nu h^\delta{}_\sigma R_{\mu\alpha\beta\gamma\delta} \\ &= R_{\nu\sigma} - h^\mu{}_\rho \varepsilon(K_{\rho\mu} K_{\sigma\nu} - K_{\sigma\mu} K_{\rho\nu}) \\ &= R_{\nu\sigma} - \varepsilon(K K_{\sigma\nu} - K_\sigma{}^\rho K_{\rho\nu}) \end{aligned}$$

$$R = h^{\nu\sigma} R_{\nu\sigma}$$

$$\begin{aligned} h^{\nu\sigma} h^\beta{}_\gamma h^\alpha{}_\nu h^\delta{}_\sigma R_{\mu\alpha\beta\gamma\delta} &= h^{\alpha\sigma} h^{\beta\delta} R_{\alpha\beta\sigma\delta} \\ &= R - \varepsilon h^{\nu\sigma} (K K_{\sigma\nu} - K_\sigma{}^\rho K_{\rho\nu}) \\ &= R - \varepsilon (K^2 - K^{\nu\sigma} K_{\nu\sigma}) \end{aligned}$$

$$\begin{aligned} R &= g^{\alpha\sigma} g^{\beta\delta} R_{\alpha\beta\sigma\delta} \\ &= (h^{\alpha\sigma} + \varepsilon n^\alpha n^\sigma)(h^{\beta\delta} + \varepsilon n^\beta n^\delta) R_{\alpha\beta\sigma\delta} \\ &\quad \underbrace{\hspace{10em}}_{\text{in eq. above}} \quad \underbrace{\hspace{10em}}_{\text{vanishes due to antisymmetry of } R} \end{aligned}$$

$$= h^{\alpha\sigma} h^{\beta\delta} R_{\alpha\beta\sigma\delta} - 2\varepsilon n^\nu n^\sigma R_{\nu\sigma}$$

$$h^{\alpha\sigma} h^{\beta\delta} R_{\alpha\beta\sigma\delta} - 2\varepsilon n^\nu n^\sigma R_{\nu\sigma} = R + \varepsilon(K^{\mu\nu} K_{\mu\nu} - K^2)$$

*) Consider now a 2-surface embedded in $\mathbb{R}^3$. The two eigenvalues of $K_\mu{}^\mu$ are the principal values of the surface, $1/R_1$ and $1/R_2$.

Prove (Gauss' Theorema Egregium) the product of the two principal radii $R_1 R_2$ is an intrinsic quantity, depending only on the induced metric on Σ and its (tangential) derivatives, but not on the normal vector and its derivatives.

Hint:

Since ambient space is flat,

$$R + \varepsilon(K^{\mu\nu} K_{\mu\nu} - K^2) = 0$$

Work in basis where K is diagonal, also note $\varepsilon =$

$$K_{11} = \kappa_1, \quad K_{22} = \kappa_2$$

$$K = \kappa_1 + \kappa_2$$

$$R = (\kappa_1 + \kappa_2)^2 - \kappa_1^2 - \kappa_2^2$$

$$= 2\kappa_1 \kappa_2$$

Hence the Gaussian curvature is intrinsic